

rate = 60 miles/hour

time = 2 hours

Distance = rate · time

$$= 60 \text{ m/h} \cdot 2 \text{ h}$$

60 miles

$$\frac{60}{1} = 60$$

$$\frac{h}{h} = 1$$

Algebra → static

calculus → change (math of change)

0

LIMITS

$f(x)$ → function of x

Formula:

$\lim_{x \rightarrow c} f(x)$

limit as x approaches c , of the function

1. Substitution $\rightarrow$ replace the a variable x with a specified number

① $\lim_{x \rightarrow 2} (x+3)$, substitute 2 for x

$$= 2+3$$

$$= 5$$

② Exponent Rule: x^2 (x multiplied by itself)
 $x \cdot x$ $(-1) \cdot (-1) = 1$ ✓

$$\lim_{x \rightarrow 5} (2x) \rightarrow \boxed{\text{algebra} = 2 \cdot x}$$

$$= 2 \cdot 5$$

$$= 10$$

③ $\lim_{x \rightarrow -1} (x^2+4) \rightarrow$ algebra rule (exponents)

$$= (-1)^2 + 4$$

$$= (-1) \cdot (-1) + 4 \rightarrow$$

$$= 1 + 4$$

$$= 5$$

④ $\lim_{x \rightarrow 3} \left(\frac{x+1}{2} \right) \longrightarrow \text{PEMDAS}$
 {
 parent
 exponent
 multiply
 divide
 add
 subtract

$$= \frac{x+1}{2}$$

$$= \frac{3+1}{2}$$

$$= \frac{4}{2}$$

$$= 2$$

algebra rule: what number times itself equals this?

⑤ $\lim_{x \rightarrow 4} \sqrt{x}$

$$= \sqrt{4}$$

$$= 2$$

4, 9, 16, 25, 36, 49, 64, 81

⑥ $\lim_{x \rightarrow 0} (5x^3 + 2x + 7)$

→ algebra rule: zero

anything multiplied by zero is zero

$$= 5x^3 + 2x + 7$$

$$= 5(0^3) + 2(0) + 7$$

$$= \underline{5 \cdot 0 \cdot 0 \cdot 0} + \underline{2 \cdot 0} + 7$$

$$= 0 + 0 + 7$$

$$= 7$$

B. Constant Rule $\rightarrow$ if there is no x , the limit is just the number itself.

$$\begin{aligned} \lim_{x \rightarrow 10} 42 \\ = 42 \end{aligned}$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \rightarrow \frac{3^2 - 9}{3 - 3} \rightarrow \frac{9 - 9}{3 - 3} \rightarrow \frac{0}{0}$$

C. Zero in denominator rule $\rightarrow$ factoring

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Difference of squares
 $(a^2 - b^2) \rightarrow (a - b) \cdot (a + b)$

$$= \frac{(x - 3) \cdot (x + 3)}{x - 3} \rightarrow \frac{x - 3}{x - 3} \cdot \frac{x + 3}{1} \rightarrow \frac{x + 3}{1}$$

$$= x + 3$$

$$\lim_{x \rightarrow 3} x + 3$$

$$\begin{aligned} &= x + 3 \\ &= 3 + 3 \rightarrow 6 \end{aligned}$$

Cancelling Rule
 $\rightarrow$ any number divided by itself is 1

$$\textcircled{9} \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \rightarrow a^2 - b^2 = (a - b)(a + b)$$

$$\underline{(x - 2)}(x + 2)$$

$$= \lim_{x \rightarrow 2} x + 2$$

$$= x + 2$$

$$= 2 + 2$$

$$= \textcircled{4}$$

$$\textcircled{10} \quad \lim_{x \rightarrow 5} \frac{2x - 10}{x - 5} \rightarrow \text{algebra rule: GCF}$$

Greater common factor

$$= \frac{2(x - 5)}{x - 5}$$

$$\textcircled{2x - 10} \rightarrow \frac{2x - 10}{2} \rightarrow \underline{(2 \cdot 1)x} - \frac{10}{2}$$

$$= \frac{2}{1} \cdot \frac{x - 5}{x - 5}$$

$$= 2 \cdot 1$$

$$= \textcircled{2}$$

